

List of Programs

- 1) Write a python program that takes in two integer input from user and print sum of those numbers.**
- 2) Write a python program that takes in two integer input from user and print difference of those numbers.**
- 3) Write a python program that takes in two integer input from user and print product of those numbers.**
- 4) Write a python program that takes in two integer input from user and print division of those numbers.**
- 5) Write a python program that takes in two integer input from user and print modulo division of those numbers.**
- 6) Write a python program to convert the temperature from Centigrade to Fahrenheit.**
- 7) Write a python program to convert the temperature from Fahrenheit to Centigrade.**
- 8) Write a python program to find the area and perimeter of rectangle.**
- 9) Write a python program to find the area and circumference of a circle.**
- 10) Write a python program to find the volume of cylinder and cone.**
- 11) Write a python program to find the area of a triangle whose sides are given.**
- 12) Write a python program to swap two numbers using third variable.**
- 13) Write a python program to swap two numbers without using third variable.**
- 14) Write a python program to find the roots of a quadratic equation.**
- 15) Write a python program to check whether the given number is even or odd.**
- 16) Write a python program to find the largest of two numbers.**
- 17) Write a python program to check whether a year is leap year or not.**
- 18) Write a python program to find the largest of three numbers.**

19) A company decides to give bonus to all its employees on Diwali. A 5% bonus on salary is given to the male worker and 10% bonus on salary to the female workers.

WAP to enter the salary of the employee and gender of the employee. If the salary of the employee is less than Rs. 10,000 then the employee gets an extra 2% bonus on salary. Calculate the bonus that has to be given to the employee and display the salary that the employee will get.

20) WAP in python to enter the marks of a student in 4 subjects. Then calculate the total and aggregate and also display the grade obtained by the student. If the student scores an aggregate $\geq 75\%$, then the grade is distinction.

If aggregate is $60 \geq$ and < 75 , then grade = 1st division

If aggregate is $50 \geq$ and < 60 , then grade = 2nd division

If aggregate is $40 \geq$ and < 50 , then grade = 3rd division else grade is fail.

21) WAP in python to determine whether the character entered is a vowel or not.

22) WAP in python that prompts the user to enter a number between 1-7 and then displays the corresponding day of the week.

23) WAP in python to take input from the user and then check whether it is a number or a character. If it is character, determine whether it is in uppercase or lowercase.

24) WAP in python to calculate the sum and average of first 10 numbers.

25) WAP in python to calculate the sum of numbers from m to n.

26) WAP in python to find whether the given number is an Armstrong number or not.

27) WAP in python to enter a decimal number. Calculate and display the binary equivalent of this number.

28) WAP in python to enter a binary number and convert it into decimal number.

29) WAP in python to calculate GCD of two numbers.

30) WAP in python to enter a number and then calculate the sum of its digits.

31) WAP in python to print the reverse of a number.

32) WAP in python to find the factorial of a number.

33) WAP in python to classify a given number as prime or composite.

34) WAP in python to find the sum of the series -- $1/1^2 + 1/2^2 + \dots + 1/n^2$.

35) WAP in python to find the sum of the series -- $1/2 + 2/3 + \dots + n/(n+1)$.

36) WAP in python to print the following pattern.

```
*  
**  
***  
****  
*****
```

37) WAP in python to print the following pattern.

```
1  
12  
123  
1234  
12345
```

38) WAP in python to print the following pattern.

```
1  
22  
333  
4444  
55555
```

39) WAP in python to print the following pattern.

```
1  
12  
123  
1234  
12345
```

40) WAP in python to print the following pattern.

```
1  
121  
12321  
1234321  
123454321
```

41) WAP in python to print the following pattern.

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

42) WAP in python to print the following pattern.

0
12
345
6789

43) WAP in python to add two integers using function.

44) WAP in python to find smaller of two numbers using function.

45) WAP in python to compare two numbers (which number is greater, which number is smaller or both are equal) using function.

46) WAP in python to swap two numbers (without using third variable) using function.

47) WAP in python using function that checks whether a passed string is palindrome or not.

48) WAP in python to print the Fibonacci series using function.

49) Write a python function to sum all the numbers in a list by:

- a) passing five different variables
- b) passing variable length argument
- c) passing a list.

50) Write a python function to check whether a number falls in a given range.

(Pass 3 variables, first the integer which is to be found, second for the starting point and third, ending point of the range).

51) Write a python function that accepts a string and calculate the number of upper case letters and lower case letters.

52) Write a python function to check whether a number is perfect or not.

(A perfect number is a positive integer that is equal to the sum of its proper positive divisors excluding the number itself.) 0

53) Create an inner function to calculate the addition in the following way

- a) Create an outer function that will accept two parameters, a and b
- b) Create an inner function inside an outer function that will calculate the addition of a and b
- c) At last, the outer function will add 5 into addition and return it

54) WAP in python using function to convert:

- a) Decimal to binary
- b) Binary to decimal

55) WAP in python that creates a list of numbers from 1-20 that is either divisible by 2 or 5.

56) WAP in python to create a list of numbers in the range 1 to 20. Then delete all the even numbers from the list and print the final list.

57) WAP in python to print index at which a particular value exists. If the value exists at multiple locations in the list, then print all the indices. Also, count the number of times that value is repeated in the list.

58) WAP in python that creates a list of words by combining the words in two individual lists. (for e.g. two lists are given l1=["hello", "python"] and l2=["world", "programming"], then the output will be another list (l3)= ['hello world', 'python programming'])

59) WAP in python to add two matrices (using nested list).

60) WAP in python that accepts different number of arguments and return sum of only positive values passed to it. (use variable length keyword)

61) WAP that has two sequences. First stores some attributes and second stores the corresponding values. Use "zip" function to get the attributes and corresponding values in the same line. (for e.g. take two list attr=["name", "age", "contact"] and val=["ABC", 20, 0123456789] and return the result 'name'- 'ABC' 'age'- 20 'contact'- 0123456789)

62) WAP in python that generates (creates) a set of prime numbers and another set of odd numbers. **Demonstrate the result of union, intersection, difference, and symmetric difference operation on these sets.**

63) WAP in python that creates a dictionary from a string containing name of states and its capital. (hint: use splitting method)

(for e.g. str="Uttar Pradesh=Lucknow, Maharashtra= Mumbai....etc.")

And output will be a dictionary containing "Uttar Pradesh": "Lucknow"....)

64) Write a python program to combine each line from first file with the corresponding line in second file and store the result in the third file.